- Use a distributed third-party storage solution to store your data in multiple availability zones: GlusterFS, DRBD, MongoDB, and so on.

The Jenkins server stores data directly on disk. To outsource the state of the virtual machine, you can't use RDS, DynamoDB, or S3; you need a file-level storage solution instead. As you've learned, an EBS volume is available only in a single availability zone, so this isn't the best fit for the problem. But do you remember EFS from chapter 9? EFS provides network file storage (over NFSv4.1) and replicates your data automatically between availability zones in a region.

To embed EFS into the Jenkins setup, as shown in listing 13.4, you have to make the following three modifications to the Multi-AZ template from the previous section:

1. Create an EFS filesystem.
2. Create EFS mount targets in each availability zone.
3. Adjust the user data to mount the EFS filesystem. Jenkins stores all its data under /var/lib/jenkins.

Listing 13.4 Storing Jenkins state on EFS

```
# [...]
FileSystem:                                                    ①
  Type: 'AWS::EFS::FileSystem'
  Properties: {}
MountTargetSecurityGroup:                                      ②
  Type: 'AWS::EC2::SecurityGroup'
  Properties:
    GroupDescription: 'EFS Mount target'
    SecurityGroupIngress:
    - FromPort: 2049                                           ③
      IpProtocol: tcp
      SourceSecurityGroupId: !Ref SecurityGroup
      ToPort: 2049
    VpcId: !Ref VPC
MountTargetA:                                                  ④
  Type: 'AWS::EFS::MountTarget'
  Properties:
    FileSystemId: !Ref FileSystem
    SecurityGroups:
    - !Ref MountTargetSecurityGroup
    SubnetId: !Ref SubnetA                                     ⑤
MountTargetB:                                                  ⑥
  Type: 'AWS::EFS::MountTarget'
```